

# Functional and differential equations

## — 2

### Functional-differential equations

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#### Theoretical minimum.

**Fact 1 (three shapes).** A functional-differential equation relates the values of  $f$  and of its derivative at *different* points. Three shapes cover almost everything: a shift,  $f'(x) = F(f(x+a))$  or  $f(x) = \int_{x-1}^x f$ ; a composition,  $f'(x) = f(f(x))$ ; and a mean value,  $f(a) - f(b) = (a-b)f'(\mu(a,b))$  for a prescribed mean  $\mu$ .

**Theorem 1 (the mean value equations).** If  $f$  is differentiable and  $f(a) - f(b) = (a-b)f'(\frac{a+b}{2})$  for all  $a, b$ , then  $f$  is a polynomial of degree at most two; for the geometric mean the answer is different and is found by first showing that  $f'$  is determined on a scaling orbit. In both cases the method is the same: fix one variable, differentiate in the other, and read the resulting ordinary differential equation.

**Theorem 2 (shifts raise regularity).** If  $f$  is continuous and  $f(x) = \int_{x-1}^x f$ , then  $f$  is  $C^\infty$ , and  $f'(x) = f(x) - f(x-1)$ ; every relation of this type is differentiated once and becomes a delay equation, whose solutions are found by substituting  $e^{\lambda x}$  and analysing the characteristic equation — which now has infinitely many roots.

**Fact 2 (compositions are rigid).** An equation such as  $f'(x) = f(f(x))$  admits almost no solutions: monotonicity and sign considerations usually force a contradiction. The standard argument compares the growth of the two sides and uses that a composition inherits monotonicity, as in problem 7 of the homework of Differentiable functions – 1.

1. (Kapacëв 4.5) Find all differentiable functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfying  $f(a) - f(b) = (a - b)f'\left(\frac{a+b}{2}\right)$  for all  $a, b \in \mathbb{R}$ .
2. (Kapacëв 4.6) Find all differentiable functions  $f: (0, +\infty) \rightarrow \mathbb{R}$  satisfying  $f(a) - f(b) = (a - b)f'(\sqrt{ab})$  for all positive  $a$  and  $b$ .
3. (ISTCiM 2024) Find all differentiable functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfying  $f(x + y) = f'(x)f(y) + f(x)f'(y)$  for all  $x, y \in \mathbb{R}$ .
4. (RMO) Find all functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that  $f^2$  is differentiable and  $f = (f^2)'$ .
5. Let  $f$  be continuous on  $[0, \infty)$  with  $f(x) = \int_{x-1}^x f(t) dt$  for all  $x \geq 1$ . Prove that  $f$  is infinitely differentiable, find the characteristic equation of the corresponding delay equation, and describe the bounded solutions. (That such an  $f$  has bounded variation is the subject of problem 5 in the homework of Definite integrals – 4.)